Answer Key - Mathematics - Undergraduate Level Assessment 25 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) 3; B) e; C) sqrt(5); D) 0
   Note: Correct option: A - 3

2. **Answer:** D
   Options: A) f(c); B) f(e^c); C) f(1/c); D) f(log c)
   Note: Correct option: D - f(log c)

3. **Answer:** C
   Options: A) 0; B) 1; C) -1; D) 12
   Note: Correct option: C - -1

4. **Answer:** C
   Options: A) 1/(2 e); B) 1/e; C) $e^2/2$; D) 2 e
   Note: Correct option: C - $e^2/2$

5. **Answer:** D
   Options: A) True, True; B) False, False; C) True, False; D) False, True
   Note: Correct option: D - False, True

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** False
    Options: A) True; B) False
    Note: LLM-generated false statement. Correct answer: '6'.

Long Form Response

11. **Answer:** Listing her first 5 times in ascending order, we get

$$86, 88, 94, 96, 97$$

Because the final median is 92 and that is between 88 and 94, the final time must also lie in this spot. Thus, we have

$$86, 88, x, 94, 96, 97$$

Because there are an even number of elements, the median is the mean of the center two. Thus, for the mean to be 92, x must be 90 *seconds*.
Note: Students should show all steps in their mathematical solution.

12. **Answer:** We use the distance formula: $\sqrt{(8-0)^2+(0-15)^2} = \sqrt{64+225} = 17. - OR -$
$Wenotethatthepoints(0, 15)$,$(8, 0), and(0, 0)formarighttrianglewithlegsoflength8and15.ThisisaPy$
Note: Students should show all steps in their mathematical solution.

End of Answer Key